

React Assignment 10 : Multi-Page Application Using React Router

Assignment Objective

Develop a multi-page web application using React that demonstrates the use of reusable components, client-side routing, and proper project structure.

Problem Statement

You are required to create a React application that consists of multiple pages. The application must use a common navigation bar component and implement routing using React Router. Each page should be implemented as a separate React component and rendered based on the route.

Functional Requirements

1. Create a React project using **Vite** or **Create React App**.
 2. Use **functional components only**.
 3. Implement client-side routing using **React Router DOM**.
 4. Create a **reusable Navbar component** that is visible on all pages.
 5. Use **Link** or **NavLink** for navigation (do not use anchor `<a>` tags).
 6. Organize the project using a proper folder structure (**components**, **pages**, etc.).
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Pages to Implement

1. **Home Page**
 - Display a welcome message.
 - Brief description of the application.
2. **About Page**
 - Display information about the application or developer.
3. **Services / Courses Page**
 - Display a list of services or courses.
 - Render the list dynamically using an array and **map()**.
4. **Contact Page**
 - Create a contact form with the following fields:

- Name
- Email
- Message
- Manage form data using the `useState` hook.
- No backend integration is required.

5. 404 Page (Not Found)

- Display a message when an invalid route is accessed.

Routing Requirements

1. Configure routing in `App.jsx`.
2. Use `BrowserRouter`, `Routes`, and `Route`.
3. Define a wildcard (*) route for the 404 page.
4. Ensure navigation occurs without page reload.

Component Requirements

1. Navbar Component

- Create as a separate reusable component.
- Must include navigation links to:
 - Home
 - About
 - Services
 - Contact

2. Footer Component

- Display basic footer information.
- Reusable across all pages.

Styling Requirements

1. Apply basic styling using:
 - CSS **or**
 - Bootstrap.
 2. Maintain consistent layout and design across all pages.
 3. Navbar should be visually clear and user-friendly.
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